UE26CY151A · ENGINEERING CHEMISTRY

SEMESTER 1 · UNIT 1

04

PROCESS MAPS

Flowcharts & Visuals

Step-by-step problem-solving flows, decision paths and comparison tables.

M1
Molecular Spectroscopy
8 topics

M2
Phase Equilibria
5 topics

M3
Computational Chemistry
1 topic

Exam weight	★★★ high · ★★☆ medium · ★☆☆ low
Flags	NOT ON CURRENT SLIDES · TRANSCRIPT ONLY · ADDED FOR CLARITY
Sources	S004·8 slide 8 of deck 004 · T009 class transcript · PDF Unit-1 slide PDF · ESA past paper

Compiled from PESU AV-summary transcripts, class slides and past ESA papers · 13 October 2026

Contents

M1 Molecular Spectroscopy

T1 EM radiation & interaction with matter ★★★

T2 Electromagnetic spectrum & regions ★★★

T3 Born–Oppenheimer approximation ★★☆

T3b Beer–Lambert law ★☆☆

T4 Rotational (microwave) spectroscopy ★★★

T5 Vibrational (IR) – SHO ★★★

T6 Vibrational – anharmonic ★★☆

T7 Electronic spectroscopy & Franck–Condon ★★★

M2 Phase Equilibria

T8 Basic terms & phase rule ★★★

T9 Derivation of phase rule ★★★

T10 One-component system: water ★★★

T11 Two-component: reduced phase rule & Pb–Ag ★★★

T12 Thermal analysis / cooling curves ★★☆

M3 Computational Chemistry

T13 Computational chemistry ★☆☆

MODULE 1

Molecular Spectroscopy

T1 · EM radiation & interaction with matter | T2 · Electromagnetic spectrum & regions | T3 · Born–Oppenheimer approximation | T3b · Beer–Lambert law | T4 · Rotational (microwave) spectroscopy | T5 · Vibrational (IR) – SHO | T6 · Vibrational – anharmonic | T7 · Electronic spectroscopy & Franck–Condon

T1 EM radiation & interaction with matter ★★★

Atomic vs molecular spectra

	Atomic	Molecular
Transition	between atomic orbitals	HOMO→LUMO + vib + rot
Appearance	line spectra	band spectra

T2 Electromagnetic spectrum & regions ★★★

Microwave vs IR vs UV-Vis

	Microwave	IR	UV-Vis
Transition	rotational	vibrational	electronic
Condition	permanent dipole	dipole changes	any molecule
Rule	$\Delta J = \pm 1$	$\Delta v = \pm 1$ (SHO)	none for v; Franck–Condon
Gives	B, I, r_o	k, $\bar{\nu}$	electronic states

Rayleigh vs Raman

	Rayleigh	Stokes	Anti-Stokes
Frequency	unchanged	lower	higher
Molecule	unchanged	gains energy	loses energy

T3 Born–Oppenheimer approximation ★★☆

No flowchart for this topic.

T3b Beer–Lambert law

★☆☆

No flowchart for this topic.

T4 Rotational (microwave) spectroscopy

★★★

F3 Spectroscopic activity checker

1. Permanent dipole? → Yes: microwave (rotational) active
2. Dipole changes during a vibration? → Yes: IR active
3. All molecules → electronic (UV-Vis) active
4. HCl: all three · CO: all three · H_2, Cl_2: electronic only · CO_2: IR (asym. stretch, bend) + electronic, not microwave · H_2O: all three · HI: all three

F4 Rotational spectrum → bond length

1. Read first line (or spacing) → B = line/2
2. $I = h / (8\pi^2 cB)$
3. $\mu = m_1m_2/(m_1+m_2)$ in kg
4. $r_o = \sqrt{(I/\mu)}$

T5 Vibrational (IR) – SHO

★★★

F5 IR line → force constant (and reverse)

1. $\bar{\nu} \rightarrow k = 4\pi^2c^2\bar{\nu}^2\mu$
2. $k \rightarrow \bar{\nu} = (1/2\pi c)\sqrt{(k/\mu)} \rightarrow ZPE = ½\bar{\nu} \rightarrow \varepsilon_v = (v+½)\bar{\nu}$

T6 Vibrational – anharmonic

★★☆

F6 Anharmonic solver

1. fundamental = $\bar{\omega}_e(1-2x_e)$; overtone = $2\bar{\omega}_e(1-3x_e)$
2. ratio overtone/(2·fund) = $(1-3x_e)/(1-2x_e)$ → x_e
3. $\bar{\omega}_e$ = fundamental/$(1-2x_e)$

SHO vs anharmonic

	SHO	Anharmonic
Potential	parabola	Morse
Levels	equal spacing	converge
Rule	Δv=±1	Δv=±1,±2,±3…
Spectrum	1 line	fundamental + overtones + hot bands

T7 Electronic spectroscopy & Franck–Condon

★★★

F7 Franck–Condon outcome

1. Compare r_e' with r_e''
2. Equal → (0,0) strongest
3. Longer → (v',0), v'>0 strongest
4. Much longer → continuum (dissociation)

MODULE 2

Phase Equilibria

T8 · Basic terms & phase rule | T9 · Derivation of phase rule | T10 · One-component system: water | T11 · Two-component: reduced phase rule & Pb–Ag | T12 · Thermal analysis / cooling curves

T8 Basic terms & phase rule

★★★

F9 Phase-rule application

1. Count P
2. Count C (minimum constituents)
3. Gas absent & constant pressure? → F=C−P+1, else F=C−P+2
4. Classify F=2 bivariant, 1 univariant, 0 invariant

T9 Derivation of phase rule

★★★

F8 Phase-rule derivation

1. Variables = P(C−1)+2
2. Constraints = C(P−1)
3. F = variables − constraints = C−P+2

T10 One-component system: water

★★★

No flowchart for this topic.

T11 Two-component: reduced phase rule & Pb–Ag

★★★

F10 **Pb–Ag cooling path**

1. Melt cools → Pb separates, melt gets richer in Ag along curve to O
2. At O (303 °C): Pb + Ag + melt, F=0
3. Below O: solid Pb + solid Ag

T12 Thermal analysis / cooling curves

★★☆

F11 **Thermal analysis → phase diagram**

1. Cool melts of several compositions
2. Record T vs time
3. Read break & halt temperatures
4. Plot vs composition → diagram

MODULE 3

Computational Chemistry

T13 · Computational chemistry

T13 Computational chemistry

★☆☆

F12 Computational methods tree

1. Classical: molecular mechanics, molecular dynamics (fast, lower accuracy)
2. Quantum: semi-empirical → DFT → ab initio (slower, more accurate)

F13 Ethylene electronic structure

1. Define geometry
2. Select basis set
3. Apply method
4. Solve equations → wavefunction/density
5. Visualise HOMO, LUMO

F14 AI-enhanced workflow

1. Quantum simulations
2. Dataset
3. Train ML model
4. Predict properties in ms

MM / MD / semi-empirical / ab initio / DFT

Method	Basis	Speed	Accuracy
MM	classical springs	fastest	lowest
MD	laws of motion	fast	low
Semi-empirical	Schrödinger, approximated	fast	medium
DFT	electron density	medium	high
Ab initio	wavefunction	slowest	highest